Wake up your rover

Name: ______________________________

Date: ____________________ **Period:** ____________________

Today I can...

Show a color to the sensor and explain the message that appears.

Words to know

Sensor: A part that detects something, such as a color.

Program: The steps the robot follows.

Input / output: The color reading goes in. A screen message comes out.

LOOK AT THE IDEA

WHITE → READY

RED → DELIVERY

OTHER → CHECK

Gather your materials

- LEGO set #45522 and its official instructions
- Connected computer or tablet
- Red, white and blue paper: one 8 cm square of each
- Two equal-height firm supports, checked by the teacher

Build a color-test station

Build with the power off. Tick each step when it is done.

1. Open LEGO's official instruction book. Find model C103 on pages 24-31. Build it with your teacher as a starting model, or use the wheeled base your teacher has already checked. The pictures in this course are project ideas, not exact LEGO assembly steps.
2. Ask the teacher to prepare a secure downward-facing color-sensor mount and clear attachment points for later lessons. Keep both wheel paths and all Stop controls clear. This is a change to the starting model and needs a real-kit test.
3. Set the robot on the two firm supports under fixed parts of its frame. Check that it cannot rock or slide. Both wheels must hang clear of the table. Do not support it by the sensor or an axle (a rod a wheel turns on).
4. Cut three paper squares, each 8 cm by 8 cm. Use red, white and blue. Put them beside the robot. Keep the same side of each paper facing the sensor.

Base reference: official LEGO C103 instructions, pages 24-31. Use the base your teacher has checked.

Sketch the setup. Label the parts you will check.

Finish the setup

5. Connect only your team’s motor and sensor in Coding Canvas. Point to the screen Stop control and the robot’s power button. Ask the teacher to check the connection.
6. Slide white paper below the sensor without touching it. Adjust the paper position until the sensor reads white. Mark that place with tape so every card goes in the same spot.
7. Build the color-to-message program below. Keep all movement stopped. Test one card at a time, then put it back in the marked spot.
8. Record all three results. Change one screen message to your team’s words and test again. Leave the sensor mount and base ready for lesson 2.

- [ ] Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Show where your robot starts and how you will stop it.

Code it. Predict. Try it.

Make the program

1. Start with all movement stopped.

2. Repeat 10 times: read the color. White shows READY; red shows DELIVERY; anything else shows CHECK. Wait 0.2 seconds before the next check.

3. Stop all movement and end. There are no driving steps in this program.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

Test it fairly

Name: ______________________

Date: ______________ **Period:** ______________

Change the paper color. Keep the light and the sensor's position the same.

A successful test

- ☐ The base is steady and both wheels stay still.
- ☐ Each color produces the message in the rule.
- ☐ Everyone can point to the sensor and the Stop controls.

Record every result

Card color	Expected message	Actual message	Wheels still?
White			
Red			
Other			

Which result stands out?

____

Change one thing. Try again.

The one thing we changed:

We kept these things the same:

Card color	Expected message	Actual message	Wheels still?
White			
Red			
Other			

How does the color make the screen message change?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.